

Euler's Step Method

Key Points

We want to solve the following differential equation for $y(x)$

$$\frac{dy}{dx} = f(x, y)$$

For a small value h , let $x_1 = x_0 + h$. The gradient at (x_0, y_0) is given by $f(x_0, y_0)$ and so a close approximation of $y_1(x_1)$ is given by

$$y_1 = y_0 + h f(x_0, y_0).$$

Repeating this inductively, we obtain

$$y_{n+1} = y_n + h f(x_n, y_n)$$

where $x_{n+1} = x_n + h$, and $y_n = f(x_n)$

Basic Skills

Q1. Rearrange the following equations into the form $\frac{dy}{dx} = f(x, y)$.

a) $y^2 \frac{dy}{dx} = 3 + x$

b) $xy = \sin(x) \frac{dy}{dx}$

c) $y = e^x \frac{dy}{dx} - x \frac{dy}{dx}$

d) $y \frac{dy}{dx} = 4 \frac{dy}{dx} - x \frac{dy}{dx}$

Q2. Let $f(x, y) = 2x^2y + y + 1$. Calculate each of the following

a) $f(1, 1)$

b) $f(0, 1)$

c) $f(-1, 1)$

d) $f(3, 7)$

Q3. Let $x_0 = 1$ and $h = 0.5$. Calculate each of the following

a) x_1

b) x_2

c) x_5

Q4. Let $y_0 = 2$, $h = 0.1$, and $f(x_0, y_0) = 20$. Calculate y_1

Competency Skills

Q5. Consider the differential equation $\frac{dy}{dx} = y + 2x^3$. Let $x_0 = 1$, $y_0 = y(1) = 3$, and $h = 0.2$.

a) Calculate x_1

b) By calculating $f(x_0, y_0)$, estimate $y_1 = y(x_1)$

c) Estimate $y_2 = y(x_2)$

d) Complete the table

e) Read off the estimate for $y(2)$ from the table

i	1	2	3	4	5
x_i					
y_i					

Q6. Consider the differential equation $\frac{dy}{dx} = \cos(x + y) + e^{xy}$, with $y(0) = 0$. Using $h = 0.25$, estimate $y(1)$.

Q7. Consider the differential equation $\sqrt{x^2 + y^2} \frac{dy}{dx} = x^2 y^2$, with $y(0) = 1$. Using $h = 1$, estimate $y(3)$.

Advanced Skills

Q8. Consider the differential equation $\frac{d^2y}{dx^2} = 2 \frac{dy}{dx} + x^2 + 5$ such that $\frac{dy}{dx}(1) = y(1) = 1$. Using $h = 0.25$, estimate $y(2)$.

Q9. a) Explain why Euler's step method gives an estimate of the function given the derivative

b) Explain why using a smaller width h makes for more accurate estimations of the function

c) Give an example of function for which a Euler step does not give a good estimation and explain why

Q10. It is proposed that instead to use our current estimate to work out the gradient at the end of a step and hence improve our current estimate by averaging the gradient. Determine the Iterative formula

Extension

Q11. Let $v = v(t)$ be the velocity of a car, and $a = a(t)$ be the acceleration. Given that $a(t) = 70(1 - e^{-vt})$ and that the car starts from rest. Estimate the speed of the car when $t = 10$.